

Formula cheat sheet

Complete placement-aptitude formula bank — 347 formulas across 40 topics, covering every quant area asked in campus tests.

01 Number System

SUM OF FIRST N NATURAL NUMBERS

$$n(n + 1) / 2$$

SUM OF FIRST N ODD NUMBERS

$$n^2$$

SUM OF FIRST N EVEN NUMBERS

$$n(n + 1)$$

SUM OF SQUARES OF FIRST N NUMBERS

$$n(n + 1)(2n + 1) / 6$$

SUM OF CUBES OF FIRST N NUMBERS

$$[n(n + 1)/2]^2$$

DIVISION IDENTITY

$$\text{Dividend} = \text{Divisor} \times \text{Quotient} + \text{Remainder}$$

NUMBER OF DIGITS IN A NUMBER N

$$\lfloor \log_{10} N \rfloor + 1$$

EVEN ± EVEN, ODD ± ODD

Always even

EVEN ± ODD

Always odd

PRODUCT OF N CONSECUTIVE INTEGERS

Always divisible by $n!$

02 HCF & LCM

CORE RELATIONSHIP

$$\text{HCF} \times \text{LCM} = \text{Product of the two numbers}$$

HCF

Product of the lowest powers of common prime factors

LCM

Product of the highest powers of all prime factors

HCF OF FRACTIONS

$$\text{HCF of numerators} / \text{LCM of denominators}$$

LCM OF FRACTIONS

$$\text{LCM of numerators} / \text{HCF of denominators}$$

CO-PRIME NUMBERS

$$\text{HCF} = 1, \text{LCM} = \text{their product}$$

LEAST NUMBER LEAVING REMAINDER R

$$\text{LCM of divisors} + r$$

LEAST NUMBER LEAVING REMAINDER X EACH TIME

$$\text{LCM} - (\text{Divisor} - x), \text{ when the gap is constant}$$

GREATEST NUMBER DIVIDING A, B, C LEAVING THE SAME REMAINDER

$$\text{HCF of the differences } (a-b), (b-c), (c-a)$$

BELLS / LIGHTS RINGING TOGETHER

$$\text{LCM of the individual intervals}$$

03 Divisibility Rules

2

Last digit is 0, 2, 4, 6 or 8

3

Sum of digits is divisible by 3

4

Last two digits form a multiple of 4

5

Last digit is 0 or 5

6

Divisible by both 2 and 3

7

Double the last digit, subtract from the rest;
result divisible by 7

8

Last three digits form a multiple of 8

9

Sum of digits is divisible by 9

10

Last digit is 0

11

Odd-place digit sum - even-place digit sum = 0
or a multiple of 11

12

Divisible by both 3 and 4

25

Last two digits are 00, 25, 50 or 75

04 Fractions & Decimals

ADDITION

$$a/b + c/d = (ad + bc) / bd$$

SUBTRACTION

$$a/b - c/d = (ad - bc) / bd$$

MULTIPLICATION

$$a/b \times c/d = ac / bd$$

DIVISION

$$a/b \div c/d = a/b \times d/c$$

COMPARING FRACTIONS

Cross-multiply, or convert each to a decimal

FRACTION TO PERCENTAGE

$$(a/b) \times 100$$

PURE RECURRING DECIMAL 0.ABAB...

$$ab / 99 \text{ (as many 9s as repeating digits)}$$

MIXED RECURRING DECIMAL 0.ABCC...

$$(abc - ab) / 900$$

HANDY EQUIVALENTS

$$1/2 = 50\%, 1/3 = 33\frac{1}{3}\%, 1/4 = 25\%, 1/5 = 20\%$$

MORE EQUIVALENTS

$$1/6 = 16\frac{2}{3}\%, 1/7 = 14\frac{2}{7}\%, 1/8 = 12\frac{1}{2}\%, 1/9 = 11\frac{1}{9}\%$$

05 Surds & Indices

PRODUCT OF POWERS

$$a^m \times a^n = a^{m+n}$$

QUOTIENT OF POWERS

$$a^m \div a^n = a^{m-n}$$

POWER OF A POWER

$$(a^m)^n = a^{mn}$$

ZERO INDEX

$$a^0 = 1 \ (a \neq 0)$$

NEGATIVE INDEX

$$a^{-n} = 1 / a^n$$

FRACTIONAL INDEX

$$a^{(m/n)} = \sqrt[n]{a^m}$$

PRODUCT OF ROOTS

$$\sqrt{a} \times \sqrt{b} = \sqrt{ab}$$

QUOTIENT OF ROOTS

$$\sqrt{a} \div \sqrt{b} = \sqrt{a/b}$$

RATIONALISING A DENOMINATOR

Multiply by the conjugate: $1/(\sqrt{a} + \sqrt{b}) \rightarrow (\sqrt{a} - \sqrt{b})/(a - b)$

COMPARING SURDS

Raise both to the LCM of their root orders

06 Unit Digit & Cyclicity

CYCLE OF 2

2, 4, 8, 6 (length 4)

CYCLE OF 3

3, 9, 7, 1 (length 4)

CYCLE OF 7

7, 9, 3, 1 (length 4)

CYCLE OF 8

8, 4, 2, 6 (length 4)

CYCLE OF 4

4, 6 (length 2)

CYCLE OF 9

9, 1 (length 2)

FIXED UNIT DIGITS

0, 1, 5 and 6 always end in themselves

METHOD

Divide the power by the cycle length; the remainder gives the position

REMAINDER 0

Take the last term of the cycle

PRODUCTS

Multiply only the unit digits, step by step

07 Remainders & Large Powers

REMAINDER IDENTITY

Dividend = Divisor $\times$ Quotient + Remainder

REMAINDER OF A PRODUCT

Multiply the individual remainders, then reduce

REMAINDER OF A SUM

Add the individual remainders, then reduce

NEGATIVE REMAINDER TRICK

Remainder of 5 mod 7 can be read as -2 to shorten work

FERMAT'S LITTLE THEOREM

$a^{p-1} \bmod p = 1$ when p is prime and a is not a multiple of p

EULER'S THEOREM

$a^{\phi(n)} \bmod n = 1$ when a and n are co-prime

EULER'S TOTIENT

$\phi(n) = n(1 - 1/p_1)(1 - 1/p_2) \dots$ over distinct primes

WILSON'S THEOREM

$(p - 1)! \bmod p = p - 1$ for prime p

CYCLIC REMAINDERS

Powers of a number modulo n repeat in a fixed cycle

08 Ratio & Proportion

SHARE OF A PART

(its ratio term / sum of terms) $\times$ total

PROPORTION

$a : b = c : d \Rightarrow ad = bc$

MEAN PROPORTIONAL BETWEEN A AND B

$\sqrt{ab}$

THIRD PROPORTIONAL TO A AND B

b^2 / a

FOURTH PROPORTIONAL TO A, B, C

bc / a

COMPONENDO & DIVIDENDO

If $a/b = c/d$, then $(a+b)/(a-b) = (c+d)/(c-d)$

DUPLICATE RATIO

$a^2 : b^2$

SUB-DUPLICATE RATIO

$\sqrt{a} : \sqrt{b}$

DIRECT PROPORTION

$a \propto b \Rightarrow a/b$ is constant

INVERSE PROPORTION

$a \propto 1/b \Rightarrow ab$ is constant

09 Combined Ratio & Partnership

COMBINING A:B AND B:C

Make B equal in both, then write A : B : C

SHORTCUT

$$A:B = a:b \text{ and } B:C = c:d \Rightarrow A:B:C = ac : bc : bd$$

A : C DIRECTLY

$$(A/B) \times (B/C)$$

SIMPLE PARTNERSHIP

Profit is shared in the ratio of the capitals

COMPOUND PARTNERSHIP

Profit ratio = capital $\times$ time invested

WORKING VS SLEEPING PARTNER

Deduct the salary or commission first, then divide the rest

10 Percentage

PERCENTAGE

$$(\text{Part} / \text{Whole}) \times 100$$

X% OF N

$$(x / 100) \times N$$

PERCENTAGE INCREASE

$$(\text{Increase} / \text{Original}) \times 100$$

PERCENTAGE DECREASE

$$(\text{Decrease} / \text{Original}) \times 100$$

VALUE AFTER AN INCREASE OF X%

$$N \times (100 + x) / 100$$

VALUE AFTER A DECREASE OF X%

$$N \times (100 - x) / 100$$

ORIGINAL VALUE FROM A NEW VALUE

$$\text{New} \times 100 / (100 \pm x)$$

A IS WHAT % OF B

$$(A / B) \times 100$$

A IS WHAT % MORE THAN B

$$((A - B) / B) \times 100$$

A IS WHAT % LESS THAN B

$$((B - A) / B) \times 100$$

11 Successive Changes & Discount

TWO SUCCESSIVE CHANGES

Net % = $a + b + ab/100$ (use a minus sign for a decrease)

INCREASE THEN EQUAL DECREASE OF X%

Net = $-x^2/100$, always a loss

SUCCESSIVE DISCOUNTS OF A% AND B%

Effective = $a + b - ab/100$

PRICE RISE WITH FIXED SPENDING

Consumption must fall by $100x/(100 + x) \%$

PRICE FALL WITH FIXED SPENDING

Consumption can rise by $100x/(100 - x) \%$

POPULATION AFTER N YEARS

$P(1 + r/100)^n$

DEPRECIATED VALUE AFTER N YEARS

$P(1 - r/100)^n$

12 Average

AVERAGE

Sum of values / Number of values

SUM

Average $\times$ Number of values

AVERAGE OF FIRST N NATURAL NUMBERS

$(n + 1)/2$

AVERAGE OF FIRST N EVEN NUMBERS

$n + 1$

AVERAGE OF FIRST N ODD NUMBERS

n

AVERAGE OF CONSECUTIVE NUMBERS

$(\text{First} + \text{Last})/2$

ADDING K TO EVERY VALUE

Average also increases by k

MULTIPLYING EVERY VALUE BY K

Average also multiplies by k

REPLACEMENT OF A MEMBER

Change in total = New value - Old value

WEIGHTED AVERAGE

$(n_1a_1 + n_2a_2) / (n_1 + n_2)$

13 Profit, Loss & Discount

PROFIT

$$SP - CP$$

LOSS

$$CP - SP$$

PROFIT %

$$(\text{Profit} / CP) \times 100$$

LOSS %

$$(Loss / CP) \times 100$$

SP FROM PROFIT %

$$CP \times (100 + P\%) / 100$$

SP FROM LOSS %

$$CP \times (100 - L\%) / 100$$

CP FROM SP

$$SP \times 100 / (100 \pm P\% \text{ or } L\%)$$

DISCOUNT

$$MP - SP$$

DISCOUNT %

$$(Discount / MP) \times 100$$

SELLING TWO ITEMS AT THE SAME PRICE WITH EQUAL % GAIN AND LOSS

$$\text{Always a net loss of } x^2/100 \%$$

DISHONEST DEALER USING A FALSE WEIGHT

$$\text{Gain \%} = (\text{Error} / (\text{True value} - \text{Error})) \times 100$$

MARKED PRICE FOR A DESIRED PROFIT

$$MP = CP \times (100 + P\%) / (100 - D\%)$$

14 Simple Interest

SIMPLE INTEREST

$$SI = P \times R \times T / 100$$

AMOUNT

$$A = P + SI$$

PRINCIPAL

$$P = SI \times 100 / (R \times T)$$

RATE

$$R = SI \times 100 / (P \times T)$$

TIME

$$T = SI \times 100 / (P \times R)$$

AMOUNT IN TERMS OF P

$$A = P(1 + RT/100)$$

TIME TO DOUBLE AT SI

$$T = 100 / R \text{ years}$$

TIME TO TRIPLE AT SI

$$T = 200 / R \text{ years}$$

15 Compound Interest

AMOUNT, COMPOUNDED ANNUALLY

$$A = P(1 + R/100)^T$$

COMPOUND INTEREST

$$CI = A - P$$

HALF-YEARLY COMPOUNDING

$$A = P(1 + R/200)^{(2T)}$$

QUARTERLY COMPOUNDING

$$A = P(1 + R/400)^{(4T)}$$

DIFFERENT RATES EACH YEAR

$$A = P(1 + r_1/100)(1 + r_2/100)...$$

CI - SI FOR 2 YEARS

$$P(R/100)^2$$

CI - SI FOR 3 YEARS

$$P(R/100)^2(300 + R)/100$$

CI FOR 2 YEARS AS A PERCENTAGE

$$2R + R^2/100 \text{ of the principal}$$

TIME TO DOUBLE AT CI (APPRX.)

$$72 / R \text{ years — the rule of 72}$$

16 Time, Speed & Distance

SPEED

$$\text{Distance} / \text{Time}$$

DISTANCE

$$\text{Speed} \times \text{Time}$$

TIME

$$\text{Distance} / \text{Speed}$$

KM/H TO M/S

$$\text{Multiply by } 5/18$$

M/S TO KM/H

$$\text{Multiply by } 18/5$$

AVERAGE SPEED

$$\text{Total distance} / \text{Total time}$$

EQUAL DISTANCES AT SPEEDS X AND Y

$$\text{Average speed} = 2xy/(x + y)$$

SAME DISTANCE, SPEEDS IN RATIO A : B

$$\text{Times taken are in ratio } b : a$$

RELATIVE SPEED, OPPOSITE DIRECTIONS

$$a + b$$

RELATIVE SPEED, SAME DIRECTION

$$a - b$$

17 Trains

CROSSING A POLE OR A MAN

$$\text{Time} = \text{Length of train} / \text{Speed}$$

CROSSING A PLATFORM OR BRIDGE

$$\text{Time} = (\text{Train} + \text{Platform length}) / \text{Speed}$$

TWO TRAINS CROSSING, OPPOSITE DIRECTIONS

$$\text{Time} = (L_1 + L_2) / (S_1 + S_2)$$

TWO TRAINS CROSSING, SAME DIRECTION

$$\text{Time} = (L_1 + L_2) / (S_1 - S_2)$$

TRAIN PASSING A MOVING MAN

Use the relative speed of train and man

TRAINS STARTING TOGETHER AND MEETING

Ratio of speeds = $\sqrt{t_2} : \sqrt{t_1}$, where t is the time after meeting

18 Boats & Streams

DOWNSTREAM SPEED

$$\text{Boat speed} + \text{Stream speed}$$

UPSTREAM SPEED

$$\text{Boat speed} - \text{Stream speed}$$

SPEED OF THE BOAT IN STILL WATER

$$(\text{Downstream} + \text{Upstream}) / 2$$

SPEED OF THE STREAM

$$(\text{Downstream} - \text{Upstream}) / 2$$

SAME DISTANCE BOTH WAYS

$$\text{Average speed} = 2 \cdot D \cdot U / (D + U)$$

ROUND TRIP TIME FOR DISTANCE D

$$d/D + d/U$$

19 Time & Work

ONE DAY'S WORK

$1 / \text{Number of days}$

COMBINED ONE DAY'S WORK

$1/a + 1/b$

DAYS WORKING TOGETHER

$ab / (a + b)$

THREE PEOPLE TOGETHER

$abc / (ab + bc + ca)$

MEN-DAYS RULE

$M_1 \times D_1 \times H_1 = M_2 \times D_2 \times H_2$

EFFICIENCY AND TIME

Inversely proportional

A IS K TIMES AS EFFICIENT AS B

A takes $1/k$ of B's time

WORK LEFT AFTER D DAYS

$1 - d \times (\text{one day's work})$

WAGES

Divided in the ratio of the work done, i.e. of efficiencies

20 Pipes & Cisterns

FILLING PIPE

Positive work: $+1/a$ per hour

EMPTYING PIPE

Negative work: $-1/b$ per hour

BOTH PIPES OPEN

Net rate = $1/a - 1/b$

TIME TO FILL WITH BOTH OPEN

$ab / (b - a)$, when $b > a$

TWO FILLING PIPES TOGETHER

$ab / (a + b)$

LEAK EMPTYING A FULL TANK

Compare the filling rate with and without the leak

21 Mixtures & Alligation

AVERAGE PRICE

Total cost / Total quantity

RULE OF ALLIGATION

(Dearer – Mean) ∶ (Mean – Cheaper) = Cheaper quantity ∶ Dearer quantity

COMPONENT IN A SOLUTION

Concentration % × Total quantity / 100

ADDING PURE SOLVENT

The solute stays the same; only the total changes

REPEATED REPLACEMENT OF X FROM A VESSEL OF V, N TIMES

Remaining = $V(1 - x/V)^n$

NEW RATIO AFTER ADDING WATER

Recompute with the unchanged component fixed

22 Ages

AGE N YEARS AGO

Present age – n

AGE N YEARS HENCE

Present age + n

RATIO PROBLEMS

Take the ages as ax and bx and form an equation

AGE DIFFERENCE

Stays constant for life

SUM AND DIFFERENCE GIVEN

Larger = (Sum + Difference)/2

RATIO NOW A:B, AFTER N YEARS C:D

Solve $(ax + n)/(bx + n) = c/d$

23 Arithmetic Progression

NTH TERM

$$a_n = a + (n - 1)d$$

SUM OF N TERMS

$$S_n = n/2 [2a + (n - 1)d]$$

SUM USING THE LAST TERM

$$S_n = n/2 (a + l)$$

NUMBER OF TERMS

$$n = (l - a)/d + 1$$

ARITHMETIC MEAN OF A AND B

$$(a + b)/2$$

MIDDLE TERM OF AN ODD-LENGTH AP

Equals the average of the whole AP

24 Geometric & Harmonic Progression

NTH TERM OF A GP

$$a_n = ar^{n-1}$$

SUM OF N TERMS OF A GP

$$S_n = a(r^n - 1)/(r - 1), r \neq 1$$

SUM OF AN INFINITE GP

$$S_{\infty} = a/(1 - r), \text{ for } |r| < 1$$

GEOMETRIC MEAN OF A AND B

$$\sqrt{ab}$$

HARMONIC PROGRESSION

The reciprocals form an AP

HARMONIC MEAN OF A AND B

$$2ab/(a + b)$$

RELATIONSHIP

$$AM \geq GM \geq HM, \text{ and } GM^2 = AM \times HM$$

25 Permutation

ARRANGEMENTS OF N DISTINCT ITEMS

$$n!$$

ARRANGEMENT OF R OUT OF N

$$nPr = n! / (n - r)!$$

WITH REPETITION ALLOWED

$$n^r$$

ITEMS NOT ALL DISTINCT

$$n! / (p! q! r!)$$

CIRCULAR ARRANGEMENT

$$(n - 1)!$$

CIRCULAR, WITH NO DISTINCTION BETWEEN CLOCKWISE AND ANTICLOCKWISE

$$(n - 1)!/2$$

ITEMS ALWAYS TOGETHER

Treat the group as one unit, then multiply by
its internal arrangements

ITEMS NEVER TOGETHER

Total arrangements - arrangements with them
together

26 Combination

SELECTION OF R OUT OF N

$$nC_r = n! / [r!(n - r)!]$$

SYMMETRY

$$nC_r = nC_{(n - r)}$$

PASCAL'S RULE

$$nC_r + nC_{(r-1)} = (n+1)C_r$$

SPECIAL VALUES

$$nC_0 = nC_n = 1 \text{ and } nC_1 = n$$

LINK WITH PERMUTATION

$$nPr = nC_r \times r!$$

TOTAL SUBSETS OF A SET OF N

$$2^n$$

HANDSHAKES AMONG N PEOPLE

$$n(n - 1)/2$$

DIAGONALS OF AN N-SIDED POLYGON

$$n(n - 3)/2$$

AT LEAST ONE SELECTION FROM N ITEMS

$$2^n - 1$$

27 Probability

PROBABILITY OF AN EVENT

$$P(E) = \text{Favourable Outcomes} / \text{Total Outcomes}$$

RANGE

$$0 \leq P(E) \leq 1$$

COMPLEMENT

$$P(\text{not } E) = 1 - P(E)$$

ADDITION RULE

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

MUTUALLY EXCLUSIVE EVENTS

$$P(A \cap B) = 0$$

INDEPENDENT EVENTS

$$P(A \cap B) = P(A) \times P(B)$$

CONDITIONAL PROBABILITY

$$P(A|B) = P(A \cap B) / P(B)$$

COINS

n coins give 2^n outcomes

DICE

One die gives 6 outcomes; two dice give 36

CARDS

52 cards: 26 red, 26 black, 13 of each suit, 4 of each rank, 12 face cards

28 Geometry — Triangles

AREA

$$1/2 \times \text{base} \times \text{height}$$

HERON'S FORMULA

$$\sqrt{[s(s-a)(s-b)(s-c)]}, \text{ where } s = (a+b+c)/2$$

EQUILATERAL TRIANGLE AREA

$$(\sqrt{3}/4) \times a^2$$

HEIGHT OF AN EQUILATERAL TRIANGLE

$$(\sqrt{3}/2) \times a$$

PYTHAGORAS THEOREM

$$a^2 + b^2 = c^2$$

ANGLE SUM

$$180^\circ$$

EXTERIOR ANGLE

Equals the sum of the two opposite interior angles

SIMILAR TRIANGLES

Sides in proportion; areas in the ratio of the squares of the sides

COMMON PYTHAGOREAN TRIPLES

3-4-5, 5-12-13, 8-15-17, 7-24-25, 9-40-41

MEDIAN TO THE HYPOTENUSE

Half the hypotenuse in a right triangle

RECTANGLE

Area = $l \times b$; Perimeter = $2(l + b)$; Diagonal = $\sqrt{l^2 + b^2}$

SQUARE

Area = a^2 ; Perimeter = $4a$; Diagonal = $a\sqrt{2}$

PARALLELOGRAM

Area = base $\times$ height

RHOMBUS

Area = $1/2 \times d_1 \times d_2$

TRAPEZIUM

Area = $1/2 \times (\text{sum of parallel sides}) \times \text{height}$

CIRCLE

Area = πr^2 ; Circumference = $2\pi r$

SECTOR

Area = $(\theta/360) \times \pi r^2$; Arc = $(\theta/360) \times 2\pi r$

ANGLE SUM OF A POLYGON

$(n - 2) \times 180^\circ$

EACH INTERIOR ANGLE OF A REGULAR POLYGON

$(n - 2) \times 180^\circ / n$

SUM OF EXTERIOR ANGLES

360° for any polygon

CUBE

Volume = a^3 ; TSA = $6a^2$; Diagonal = $a\sqrt{3}$

CUBOID

Volume = lbh ; TSA = $2(lb + bh + hl)$; Diagonal = $\sqrt{l^2 + b^2 + h^2}$

CYLINDER

Volume = $\pi r^2 h$; CSA = $2\pi rh$; TSA = $2\pi r(r + h)$

CONE

Volume = $(1/3)\pi r^2 h$; CSA = πrl ; TSA = $\pi r(r + l)$

SLANT HEIGHT OF A CONE

$l = \sqrt{r^2 + h^2}$

SPHERE

Volume = $(4/3)\pi r^3$; Surface area = $4\pi r^2$

HEMISPHERE

Volume = $(2/3)\pi r^3$; TSA = $3\pi r^2$

CAPACITY

$1 \text{ m}^3 = 1000 \text{ litres}$; $1 \text{ litre} = 1000 \text{ cm}^3$

31 Coordinate Geometry

DISTANCE BETWEEN TWO POINTS

$$\sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$$

MIDPOINT

$$((x_1 + x_2)/2, (y_1 + y_2)/2)$$

SECTION FORMULA (INTERNAL)

$$((mx_2 + nx_1)/(m+n), (my_2 + ny_1)/(m+n))$$

SLOPE

$$m = (y_2 - y_1)/(x_2 - x_1)$$

SLOPE-INTERCEPT FORM

$$y = mx + c$$

POINT-SLOPE FORM

$$y - y_1 = m(x - x_1)$$

PARALLEL LINES

Equal slopes

PERPENDICULAR LINES

$$m_1 \times m_2 = -1$$

AREA OF A TRIANGLE FROM VERTICES

$$1/2 |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

CENTROID

$$((x_1 + x_2 + x_3)/3, (y_1 + y_2 + y_3)/3)$$

32 Heights & Distances

BASIC RATIOS

$\sin \theta = \text{opposite/hypotenuse}$, $\cos \theta =$

$\text{adjacent/hypotenuse}$, $\tan \theta =$

opposite/adjacent

STANDARD VALUES OF TAN

$$\tan 30^\circ = 1/\sqrt{3}, \tan 45^\circ = 1, \tan 60^\circ = \sqrt{3}$$

STANDARD VALUES OF SIN

$$\sin 30^\circ = 1/2, \sin 45^\circ = 1/\sqrt{2}, \sin 60^\circ = \sqrt{3}/2$$

IDENTITY

$$\sin^2 \theta + \cos^2 \theta = 1$$

ANGLE OF ELEVATION

Measured upward from the horizontal

ANGLE OF DEPRESSION

Measured downward from the horizontal

HEIGHT AT 45°

Height equals the horizontal distance

SHADOW LENGTH

Height / $\tan \theta$

33 Algebra Identities

SQUARE OF A SUM

$$(a + b)^2 = a^2 + 2ab + b^2$$

SQUARE OF A DIFFERENCE

$$(a - b)^2 = a^2 - 2ab + b^2$$

DIFFERENCE OF SQUARES

$$a^2 - b^2 = (a + b)(a - b)$$

CUBE OF A SUM

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

CUBE OF A DIFFERENCE

$$(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

SUM OF CUBES

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

DIFFERENCE OF CUBES

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

THREE-TERM SQUARE

$$(a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

USEFUL RESULT

$$a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2+b^2+c^2 - ab - bc - ca)$$

RECIPROCAL IDENTITY

$$x^2 + 1/x^2 = (x + 1/x)^2 - 2$$

34 Linear & Quadratic Equations

LINEAR EQUATION

$$ax + b = 0 \Rightarrow x = -b/a$$

TWO VARIABLES

Solve by substitution or elimination

UNIQUE SOLUTION

$$a_1/a_2 \neq b_1/b_2$$

NO SOLUTION

$$a_1/a_2 = b_1/b_2 \neq c_1/c_2$$

INFINITE SOLUTIONS

$$a_1/a_2 = b_1/b_2 = c_1/c_2$$

QUADRATIC FORMULA

$$x = [-b \pm \sqrt{b^2 - 4ac}] / 2a$$

DISCRIMINANT

$$D = b^2 - 4ac$$

NATURE OF ROOTS

$D > 0$ real and distinct, $D = 0$ equal, $D < 0$ imaginary

SUM OF ROOTS

$$-b/a$$

PRODUCT OF ROOTS

$$c/a$$

BUILDING A QUADRATIC

$$x^2 - (\text{sum})x + (\text{product}) = 0$$

35 Logarithms

DEFINITION

$$\log_a N = x \Leftrightarrow a^x = N$$

PRODUCT RULE

$$\log(mn) = \log m + \log n$$

QUOTIENT RULE

$$\log(m/n) = \log m - \log n$$

POWER RULE

$$\log(m^n) = n \log m$$

BASE CHANGE

$$\log_a N = \log_b N / \log_b a$$

LOG OF THE BASE

$$\log_a a = 1$$

LOG OF ONE

$$\log_a 1 = 0$$

RECIPROCAL

$$\log_a b = 1 / \log_b a$$

COMMON VALUES

$$\log_{10} 2 \approx 0.3010, \log_{10} 3 \approx 0.4771$$

36 Set Theory & Venn Diagrams

TWO SETS

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

THREE SETS

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$$

ONLY A

$$n(A) - n(A \cap B) - n(A \cap C) + n(A \cap B \cap C)$$

EXACTLY TWO SETS

$$n(A \cap B) + n(B \cap C) + n(C \cap A) - 3n(A \cap B \cap C)$$

NEITHER OF THE SETS

$$\text{Total} - n(A \cup B)$$

NUMBER OF SUBSETS

$$2^n$$

NUMBER OF PROPER SUBSETS

$$2^n - 1$$

37 Clocks

ANGLE COVERED BY THE HOUR HAND

0.5° per minute

ANGLE COVERED BY THE MINUTE HAND

6° per minute

ANGLE BETWEEN THE HANDS

$|30H - 5.5M|$ degrees

HANDS COINCIDE

Every $65 \frac{5}{11}$ minutes, 22 times a day

HANDS AT RIGHT ANGLES

Twice every hour, 44 times a day

HANDS IN A STRAIGHT LINE, OPPOSITE

Every $65 \frac{5}{11}$ minutes, 22 times a day

GAIN OR LOSS OF A FAULTY CLOCK

Compare its 24-hour movement with the correct clock

38 Calendars

ODD DAYS

Remainder when the number of days is divided by 7

ORDINARY YEAR

$365 \text{ days} = 52 \text{ weeks} + 1 \text{ odd day}$

LEAP YEAR

$366 \text{ days} = 52 \text{ weeks} + 2 \text{ odd days}$

LEAP YEAR RULE

Divisible by 4; century years must be divisible by 400

100 YEARS

5 odd days

200 / 300 / 400 YEARS

3 / 1 / 0 odd days

DAY CODES

0 = Sunday, 1 = Monday, ... 6 = Saturday

SAME CALENDAR REPEATS

After 6, 11 or 28 years, depending on leap years

39 Data Interpretation

PERCENTAGE SHARE

$$(\text{Part} / \text{Total}) \times 100$$

PERCENTAGE CHANGE

$$((\text{New} - \text{Old}) / \text{Old}) \times 100$$

RATIO COMPARISON

Convert to decimals before ranking

AVERAGE ACROSS A TABLE

Total of the row or column / number of entries

PIE CHART

$$\text{Total} = 360^\circ; \text{each } 1\% = 3.6^\circ$$

VALUE FROM A PIE CHART

$$(\text{Sector angle} / 360) \times \text{Total value}$$

COMPOUND GROWTH OVER N PERIODS

$$\text{Final} = \text{Initial} \times (1 + r)^n$$

40 Calculation Shortcuts

SQUARES TO MEMORISE

$$1 \text{ to } 30, \text{ and } 25^2 = 625, 35^2 = 1225, 45^2 = 2025$$

SQUARE OF A NUMBER ENDING IN 5

$$n5^2 = n(n+1) \text{ followed by } 25$$

MULTIPLY BY 11

Add adjacent digits and place between the outer digits

MULTIPLY BY 5

Halve the number, then multiply by 10

MULTIPLY BY 25

Divide by 4, then multiply by 100

NUMBERS NEAR 100

$$(100-a)(100-b) = (100 - a - b) \text{ then append } ab$$

PERCENTAGE SWAP

$$x\% \text{ of } y = y\% \text{ of } x$$

FRACTION TO PERCENTAGE SPEED

Learn $1/2$ to $1/12$ as percentages

CUBES TO MEMORISE

1 to 15

APPROXIMATION

Round to convenient figures first, refine only if the options are close

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